

ARM-450

A 450 mm printed 6-DOF manipulator with quick-change end effectors

Design · analysis · pre-print verification

1365 g

arm mass, of 1450 g

492 mm

reach with a gripper

27 / 27

verification checks

0

parts printed yet

Everything in this deck is analytical or simulated.

No part has been printed or measured — and that is deliberate:
a prediction recorded before the test is evidence, one recorded after it is not.

Two demonstrations, two different problems

and they fail for opposite reasons

- **Sinusoidal trace — a PRECISION problem.** A tool follows a sine on a vertical board. The traced curve is a permanent, visible record of repeatability, and it exercises five joints at once.
- **Docking — a CAPTURE TOLERANCE problem.** A probe enters a passive port and latches. The interesting engineering is that it must succeed while the arm is far less accurate than the interface naively requires.
- The arm is accurate to about 2.2 mm. The docking cone accepts ± 9.75 mm.
- **Docking was made feasible by geometry, not by making the arm accurate. That is the transferable idea in this project.**

Requirements

and where each one stands today

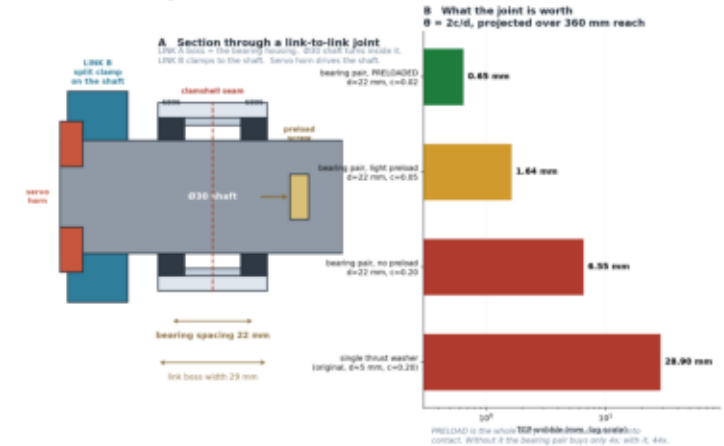
requirement	as built	status
R1 ≤ 450 mm chain	450.0 mm exactly	MET
R2 ≤ 1450 g (hard)	1365 g, 85 g spare	MET
R3 300 g payload at the tool	worst fitted case 75 g	MET
R4 6 DOF wrist	J4 roll · J5 pitch · J6 roll	MET
R5 consumer FDM printable	0.6 mm nozzle, no support on load paths	MET
R6 tool change without tools	3-lug bayonet + 1 thumbscrew	MET
R7 docking without a 7th actuator	latch driven by J6	MET
R8 visibly accurate trace	0.055 mm rms simulated at the tip	SIM ONLY
R1 constrains the ARM. With a gripper the system reaches 492 mm — raised, and accepted as a design decision, now a checked constant so a later, longer tool cannot quietly exceed it.		

One joint interface, all six axes

the decision everything else compounds from

- **Ø42 pocket · 6806 bearing · Ø30 shaft — everywhere.**
- One fit tolerance to characterise, not six — a 0.9 h coupon gates ≈ 19 h of parts.
- One spare part number, whichever joint fails.
- Clamp, horn adapter and shaft are common parts, $\times 6$.
- **Cost: the wrist carries more bearing than it needs.**
- Measured, not assumed: a 4 mm-wide 6706 there saved 24 g and cost NO stiffness — moment stiffness goes as bearing SPACING squared, and is independent of bearing width.

ARM-450 — the joint: how the links actually move

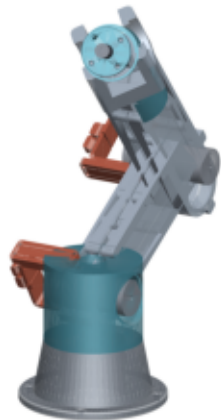


The arm

1365 g · 450 mm · 21 printed parts

ARM-450 · 6-DOF full assembly · 450 mm, 360 mm reach

J1	+25°	J2	+58°	J3	-96°
J4	+80°	J5	-38°	J6	+60°



- **Clamshell links — tongue and groove halves bolted on the neutral axis, closing a 47.5×29 box section.**
- An open C-section of the same envelope has $\approx 100\times$ less torsional stiffness — and the payload hangs off-axis.
- **Integrated wrist, not generic modules.**
- J5 is -91° to $+49^\circ$, not $\pm 110^\circ$: a mesh sweep found J4 striking J6 over the excluded range.

The kinematics, checked three independent ways

modified (Craig) DH · velocity-propagation Jacobian · ikpy

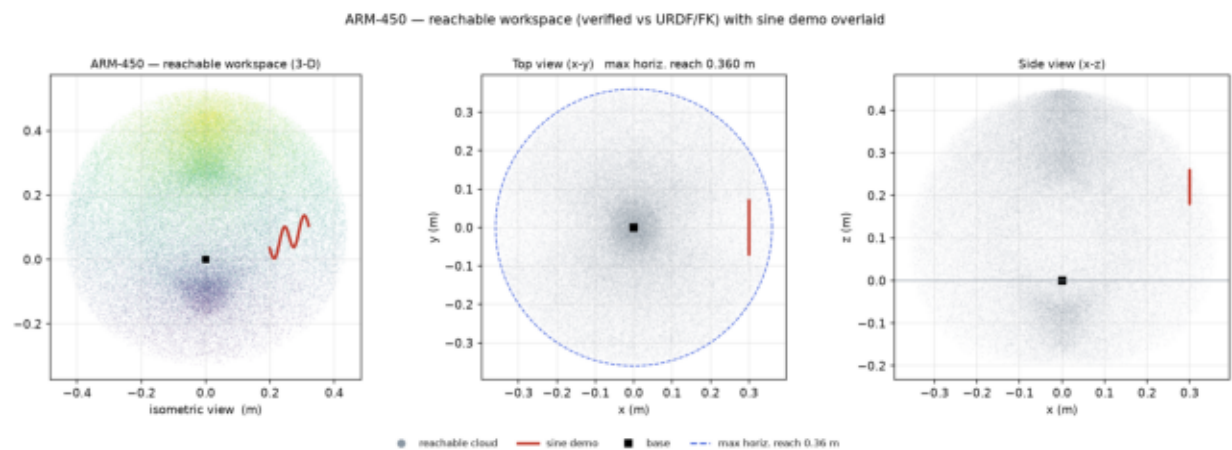
check	worst over 200 poses
FK: Craig DH == URDF chain	1.96e-13 mm
FK: Craig DH == ikpy	2.17e-13 mm
FK: URDF chain == ikpy	1.14e-13 mm
tool axis direction	8.95e-16
Jacobian: propagation == finite diff	1.81e-05
IK, Craig DH + propagated Jacobian	1.6e-11 mm
IK, ikpy	0.000 mm

Agreement between the DH table and the URDF proves only that the table copies the URDF faithfully — INCLUDING any error the URDF already had. A third route sharing no code closes that: three implementations cannot agree on one mistake.

d4 = 181, not 119. Using the obvious link length put the tool 193 mm out — caught by the URDF comparison on the first run.

Workspace

200 000-pose Monte-Carlo FK inside the URDF limits



fitted	reach	swept
bare face	450 mm	188 L
pen	480 mm	239 L
gripper	492 mm	262 L
dock probe	486 mm	251 L

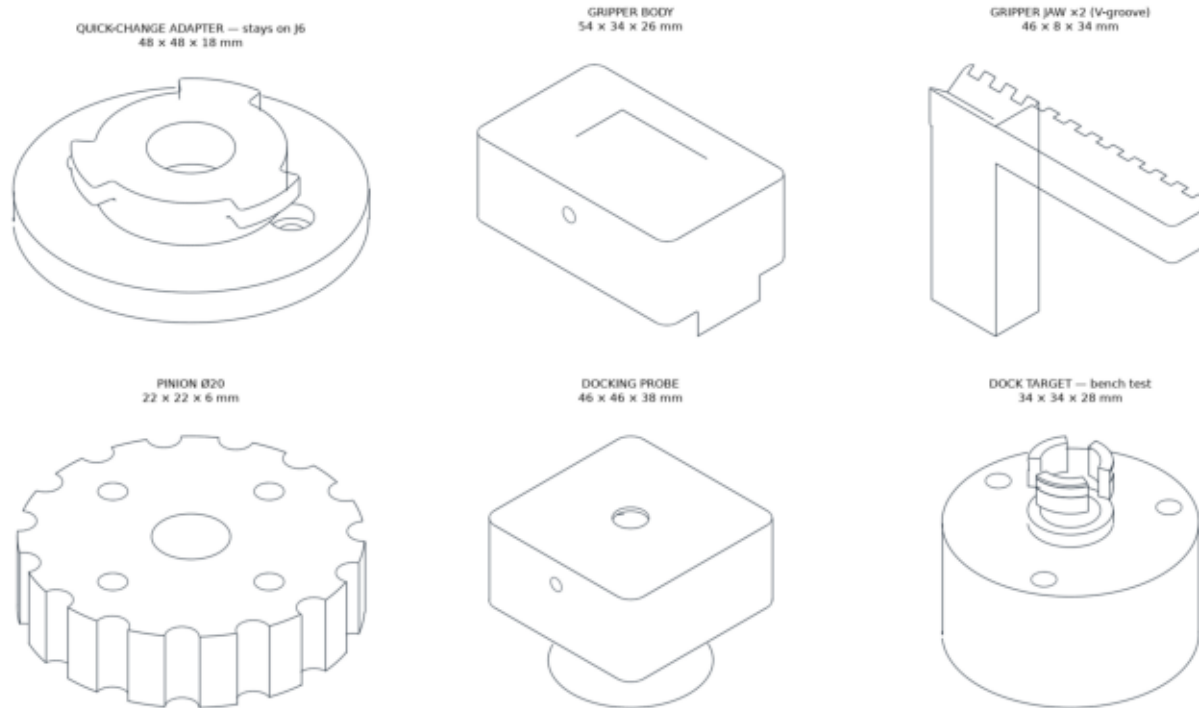
A gripper adds 39 % of swept volume for 42 mm of extension — volume grows roughly as the cube of reach.

Reach past the flange is MEASURED off the assembly, not typed in. The dock figure was stale by 7 mm until it was.

Quick-change end effectors

push on, twist 30°, nip one screw — off in about five seconds

ARM-450 — quick-change end effectors · three-lug bayonet, one thumbscrew



- **Three-lug bayonet + one M3 thumbscrew.**
- Bolts were rejected: the J6 face has only THREE holes — the servo pocket ate the fourth quadrant — and three blind M3s under a wrist is the wrong answer to a tool-change problem.
- **Gripper: parallel jaws, rack and pinion, one 9 g SG90.**
- **Dock probe: no actuator at all — J6 turns the latch.**

Docking: capture tolerance is the design driver

not accuracy — and that is the point

± 9.75 mm

≈ 2.2 mm

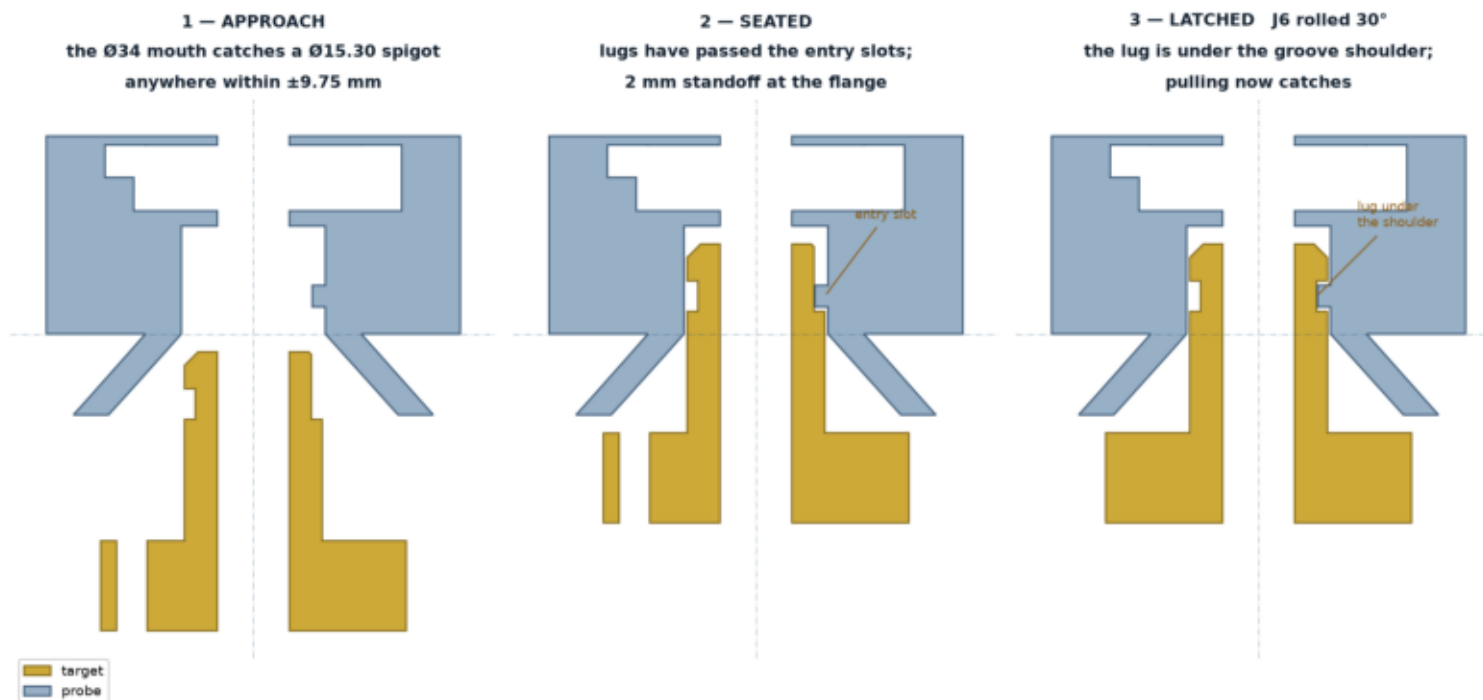
4x

measured capture

arm precision

margin

MASCOT DOCKING LATCH — section through a lug centre



Ø34 mouth · Ø16 throat · cone 9 deep · spigot 21 · groove 1.20 deep leaving 2.45 mm of wall · measured capture ± 9.75 mm · no seventh actuator

Measured by sweeping the exported insert mesh against the exported probe mesh. J6 rolled 30° latched. Reverse to unlock. They foul — not subtracted from two diameters. J6 has $\pm 175^\circ$ and does nothing during a docking approach, so the motion is free.

Verification: interrogate the geometry, not the parameters

the single most important decision in the method

- Every check reads the EXPORTED STEP or mesh — never the numbers used to author it.
- **An earlier gate checked that BEARING_POCKET_D == 42.0 and passed 55/55 while three bearing pockets did not physically exist.**
- A hollowing operation had removed the material the pocket was cut into. Subtracting a Ø42 pocket from an already-empty region succeeds silently and changes nothing.

"A boolean that removes nothing does not fail. It returns the same solid, and every step downstream — render, drawing, mass, collision — treats the result as correct."

Eight staged gates

each a real analysis, each with its own criterion

stage	question it answers	criteria
1 Design check	do the features physically exist?	STEP B-rep audit
2 FEA	does anything yield?	p99 von Mises SF ≥ 3
3 Creep + fatigue	does it fail slowly?	< 0.5 %/y, > 1e8 cycles
4 Stiffness	does the tool stay put?	compliance ≤ 0.30 mm
5 URDF	does the model match the machine?	parses, mass, meshes
6 Path tracing	does the real trajectory work?	0 collisions
7 Simulation	will ROS work on hardware?	ramp, topic, rate
8 End effectors	do the tools fit, does it latch?	29 checks + a mate

27

checks

8

stages

0

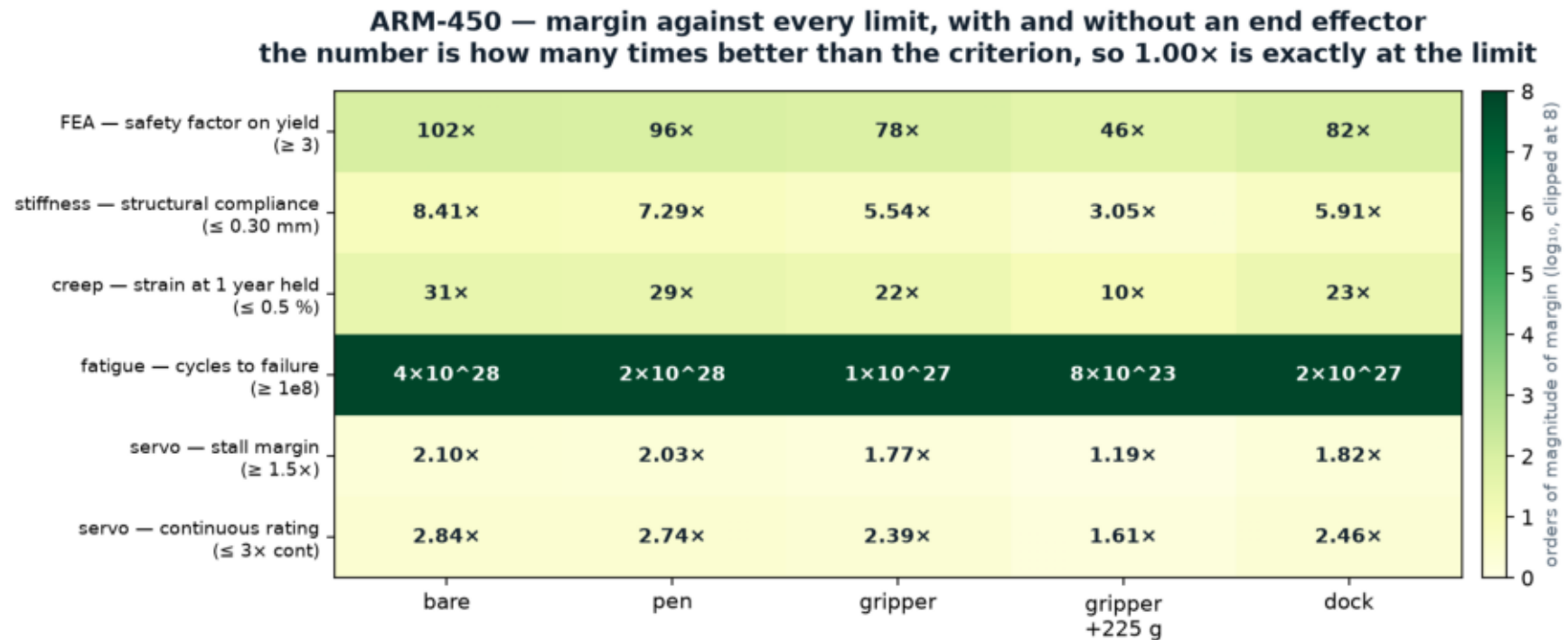
failures

GO

for printing

Every load case, with and without an end effector

because a tool is not just mass — it is a lever arm



Every cell passes. The two that are merely comfortable rather than enormous — servo stall margin and continuous rating in the gripper+225 g case — are the ones worth watching on hardware; the structural margins are so large because this arm is stiffness-driven and servo-limited, not strength-driven.

75 g on a 42 mm stick is not the load case 75 g at the flange is. The suite runs from one driver for all five configurations, so the "with gripper" answer cannot be a different vintage from "bare".

Results across five payload configurations

30 of 30 criteria pass

configuration	J2 torque	SF	compliance	servo margin	
bare	1.55 N·m	305×	0.036 mm	3.16×	PASS
pen	1.61 N·m	288×	0.041 mm	3.04×	PASS
gripper	1.85 N·m	234×	0.054 mm	2.65×	PASS
gripper + 225 g	2.74 N·m	137×	0.098 mm	1.79×	PASS
dock	1.79 N·m	245×	0.051 mm	2.73×	PASS

Structural margins of 46-102× are not a claim of excellence. They say the arm is **STIFFNESS-DRIVEN AND SERVO-LIMITED**, not strength-driven.

The two margins that are merely comfortable — servo stall 1.19×

and continuous 1.61×

in the worst case — are the ones to watch on hardware.

The precision budget has one term in it

and it is not the printed structure

configuration	structural	0.5° backlash	combined
bare	0.036 mm	1.88 mm	1.88 mm
gripper	0.054 mm	2.24 mm	2.24 mm
gripper + 225 g	0.098 mm	2.24 mm	2.24 mm

Backlash exceeds structural deflection by $\approx 20\times$, in EVERY configuration.

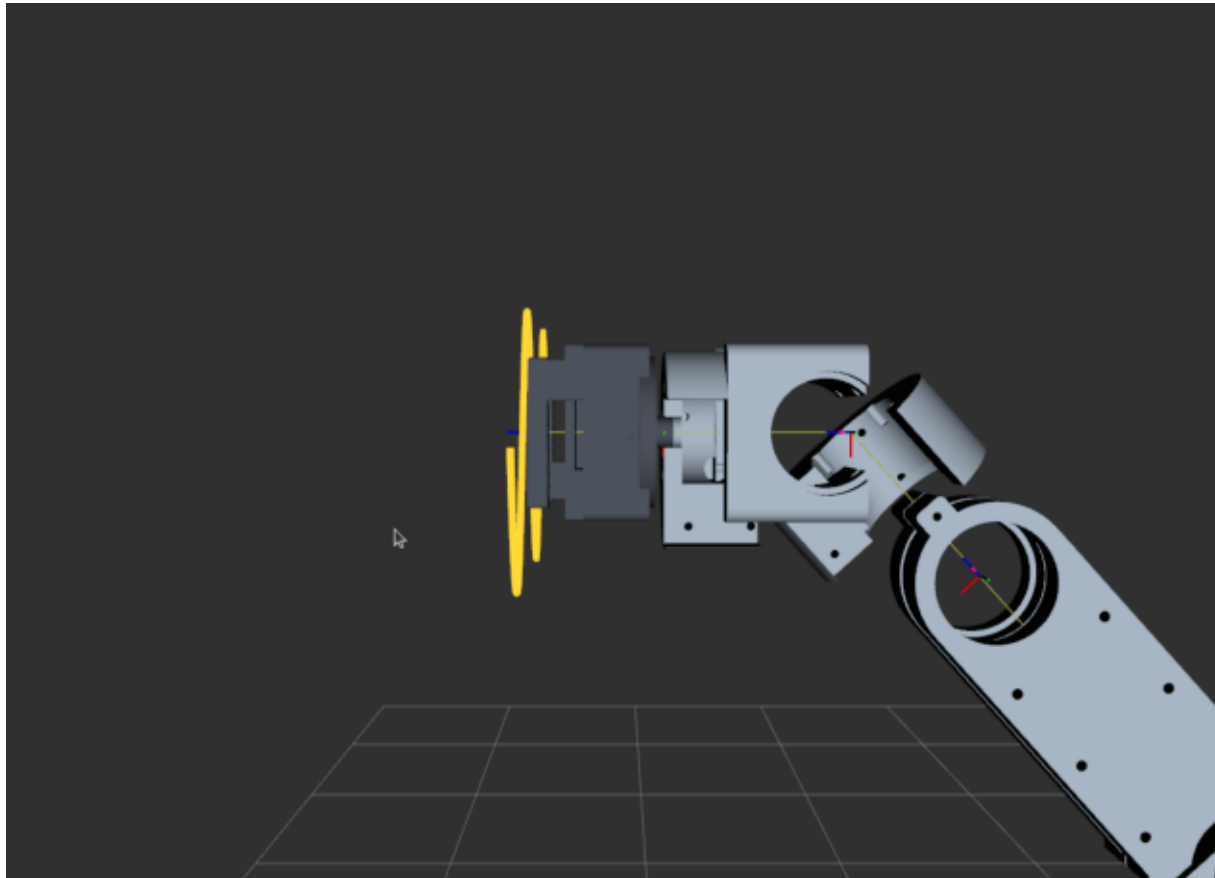
No change to the printed parts moves the number.

Accuracy work belongs in joint-side feedback, not in the arm.

This also means the conclusion is invariant to whether a tool is fitted — which is worth knowing before spending print time trying to stiffen anything.

Simulation — the trajectory that will actually run

IK solved offline, replayed by a ROS 2 node



- **0.014 mm rms at the flange, 0.055 mm at the tip.**
- **0 collisions in 240 waypoints on exact meshes; 0/60 with either tool fitted.**
- Largest joint step 2.09°.
- **Marker on the fingertips — verified at 0.00 mm from the furthest point of the tool mesh.**
- **Board goes at 342 mm, not 300: the path drives the FLANGE.**

What the gate caught

13 defects that survived visual inspection

defect	class of error
Three bearing pockets absent	a boolean that removed nothing
Dock probe exported as 4 solids	latch lugs floating 5.5 mm clear
Target groove severed its own spigot	3.30 mm cut in a 1.65 mm wall
Cone deeper than the spigot was long	the two could never have mated
Rack pitch line 1.20 mm off the PCD	the gear mesh never touched
Steel shaft quoted at 93 g, was 222 g	per-unit figure never multiplied
Sine path grazing the base pedestal	accuracy is not clearance
The URDF had no tool link at all	verified in one model, shown from another
The pattern across all of them: NOTHING FAILED LOUDLY. Each produced a plausible artefact — a render, a number, a passing check — that was wrong.	
Sine traced by the flange, not the tip	a 42 mm error on hardware

Is it space grade?

true in one specific sense, false in every other

What survives a move to flight hardware, and what does not			
SUBSYSTEM	ARM-450 (this build)	SPACE-GRADE EQUIVALENT	
Kinematics & DOF layout	6R, 450 mm, offset-free wrist	identical	
Joint architecture	Ø42 pocket - 6806 - Ø30 shaft, one part number throughout	same scheme, flight bearings	
Fastening scheme	M3 steel, heat-set brass inserts	same layout; locked, staked, vented	
Structural material	PLA+CF, E ≈ 7 GPa, Tg 60 °C	Al 6061-T6 or CFRP, E 70 GPa	
Mass	1.365 kg printed	≈1.13 kg in Al 6061, ~10x stiffer	
Actuators	ST3215/3250 hobby serial servos, plastic gear train	BLDC + harmonic drive, vacuum-rated, rad-tolerant	
Feedback	servo-internal pot, no joint encoder	joint-side resolver / optical encoder	
Precision	≈2.2 mm, backlash-dominated	< 50 µm, encoder-closed	
Lubrication	wet lithium grease in 6806	dry film MoS ₂ / PFPE, vacuum-stable	
Thermal range	0-50 °C; Tg 60 °C is the wall	-120 to +120 °C cycling, CTE-matched	
Vacuum compatibility	none — PLA outgasses, trapped volumes unvented	TML < 1 %, CVCM < 0.1 %, vented	
Radiation	none	TID-rated parts, latch-up immune	
Qualification	analysis + bench only	vibration, shock, thermal-vac, EMC	
Redundancy	single string	dual-wound motors, redundant sensing	
Docking interface	3-lug bayonet, J6-driven, ±9.75 mm capture	same principle; softer capture, latch preload, fluid seal	
Cost	consumer printer + hobby servos	3-4 orders of magnitude higher	
survives the transition unchanged survives in principle, needs rework does not survive — must be replaced			

3 survive unchanged

2 need rework

11 must be replaced

SURVIVES
the kinematics, the joint architecture,
the fastening scheme.

DOES NOT
the material (Tg 60 °C, outgasses),
the actuators (plastic gears, wet
grease, no rad tolerance), vacuum
compatibility, qualification.

The honest summary

worth saying out loud

ARM-450 is a KINEMATIC AND ARCHITECTURAL PROTOTYPE of a space manipulator, and a FUNCTIONAL DEMONSTRATOR of a docking interface.

It is not a flight article, and no part of it is on a path to becoming one without replacing the material, the actuators and the electronics.

What DOES transfer: the geometry, the joint scheme, the docking principle — and the verification method, which is grade-independent.

Re-made in Al 6061-T6 the same geometry masses ≈ 1.13 kg — LIGHTER than the printed version — and is about 10× stiffer. The material swap is not a compromise; it is an improvement that costs money.

What is NOT verified

written before the test, so it counts as a prediction

measurement	prediction	why it matters
Fit coupon — Ø42 bore	firm thumb press	gates ≈19 h of parts
Servo backlash	0.3–1.0° per joint	the dominant error term
Sine trace accuracy	1–3 mm rms	vs 0.055 mm simulated
Docking latch, bench	latches + releases	needs no arm, no fuel
Static deflection, 300 g	0.098 mm	falsified if > 0.30 mm
J2 torque and case temp	2.74 N·m, 44 °C	falsified if > 60 °C
Assembled mass	1365 ± 40 g	falsified if > 1450 g

Each has a stated FALSIFICATION condition, not just a target. §9 of the paper has the blank spaces for the measured numbers.

Limitations

stated plainly, because a verification document that does not, is not one

- **No hardware. Every number is analytical or simulated.**
- **FEA is part-wise, not assembly-level — joint stiffness, bolt preload and contact are not modelled.**
- **Backlash is assumed, not measured — and it dominates the error budget, so the headline precision figure is the least evidenced number in the work.**
- Creep uses a Findley fit two orders of magnitude below its calibration range.
- Print anisotropy is not modelled; the FEA is isotropic.
- The docking mate is rigid-body geometry — no friction, compliance or misalignment dynamics.

Conclusions

- A 450 mm printed 6-DOF arm at 1365 g with a 300 g allowance is achievable — 46-102× on yield, 0.098 mm compliance in the worst configuration.
- The arm is servo-limited, not structure-limited. Backlash beats structural deflection $\approx 20\times$, in every payload configuration.
- Docking is feasible anyway, because capture tolerance was designed as geometry. Making the interface forgiving is cheaper than making the arm accurate.
- One joint interface across six axes compounds into fewer tolerances, one spare, and a one-coupon qualification path.
- Verification must interrogate exported geometry. 13 defects survived visual inspection and passing parameter checks.

Next: print the fit coupon (0.9 h), then the arm (≈ 21 h), then measure everything in §9.